

SHARP WASSERSTEIN PROPAGATION OF CHAOS FROM CORRELATED INITIAL DATA

ABSTRACT. We study quantitative propagation of chaos for mean field diffusions with bounded smooth interaction kernels. If the squared quadratic Wasserstein distance between each initial k -particle marginal and its limiting product law is bounded by $C_0(k/N)^2$, then the same order holds uniformly on every finite time interval. The initial particle law may be correlated and singular, and the limiting law need not satisfy a transport or logarithmic Sobolev inequality. The proof combines an entropy–cost regularization estimate with a hierarchy for weighted negative Sobolev norms. The essential estimate compares the energy of a k -particle tangent with the nonnegative increment between the $(k+1)$ - and k -particle energies. Integrating this estimate along a semigroup interpolation yields the Wasserstein bound. We give an explicit admissible constant depending only on the interaction and its first derivatives, and a Gaussian example showing that the order $(k/N)^2$ is optimal for this class of initial data.

1. INTRODUCTION

Consider the system of interacting diffusions

$$dX_i^N(t) = \frac{1}{N} \sum_{j=1}^N b(X_i^N(t), X_j^N(t)) dt + \sqrt{2} dW_i(t), \quad i = 1, \dots, N, \quad (1.1)$$

where $W_1, \dots, W_N$ are independent standard Brownian motions in $\mathbb{R}^d$ and $b : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is bounded and smooth. The initial positions are exchangeable and independent of the Brownian motions. The associated McKean–Vlasov equation is

$$d\bar{X}(t) = B_t(\bar{X}(t)) dt + \sqrt{2} dW(t), \quad B_t(x) = \int_{\mathbb{R}^d} b(x, y) \mu_t(dy), \quad \mu_t = \text{Law}(\bar{X}(t)). \quad (1.2)$$

Write $P_{N,t}$ for the joint law of the particle system and $P_{N,t}^{(k)}$ for its first k -particle marginal. Propagation of chaos concerns the convergence of $P_{N,t}^{(k)}$ to $\mu_t^{\otimes k}$ as $N \rightarrow \infty$, assuming a corresponding convergence at time zero. This formulation goes back to Kac and McKean [6, 9]; see Sznitman [10] for a systematic treatment and Chaintron and Diez [3] for a survey.

For regular interactions, synchronous coupling gives quantitative bounds in Wasserstein distance. A global estimate for the full particle system, however, typically yields only a bound of order k/N for the squared distance of the k -particle marginal. The same loss occurs when one deduces local relative entropy estimates from a bound on the full N -particle entropy. The latter approach has the advantage of accommodating interactions of low regularity; see, for example, Jabin and Wang [5].

Lacker [7] developed a local approach based on the BBGKY hierarchy and obtained the sharp order $(k/N)^2$ for the relative entropy, under a corresponding initial entropy bound and an interaction-adapted transport inequality. For bounded interactions, the required inequality follows from Pinsker’s inequality. When the limiting law also satisfies a quadratic transport inequality, the entropy estimate gives the same order for the squared Wasserstein distance. Lacker and Le Flem [8] obtained time-uniform versions under additional logarithmic Sobolev and noise-strength assumptions.

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Here we start with Wasserstein information alone. More precisely, we assume that, for one constant C_0 and every marginal level,

$$W_2^2(P_{N,0}^{(k)}, \mu_0^{\otimes k}) \leq C_0 \frac{k^2}{N^2}, \quad 1 \leq k \leq N. \quad (1.3)$$

We prove that this estimate propagates on every finite time interval. In particular, we do not assume that the initial particle law has finite relative entropy with respect to $\mu_0^{\otimes N}$. Nor do we impose a transport inequality on μ_0 or on its evolution. The initial laws may be singular, subject only to finite second moments. The result concerns time marginals and bounded smooth interactions; it is not a path-space or a uniform-in-time estimate.

1.1. Outline of the argument. The first step is to evolve the initial particle law with the independent reference transitions from (1.2). If $Q_{r,s}$ denotes the one-particle transition family for the drift B , set

$$R_s^N = P_{N,0} Q_{0,s}^{\otimes N}. \quad (1.4)$$

Although the transitions act independently, R_s^N retains correlations from $P_{N,0}$. An entropy–cost inequality regularizes (1.3) and gives

$$H(R_s^{N,(k)} \mid \mu_s^{\otimes k}) \leq C_0 \beta(s) \frac{k^2}{N^2}, \quad \beta(s) = O(s^{-1}) \quad \text{as } s \downarrow 0. \quad (1.5)$$

This estimate is used to control the difference between the reference and interacting generators, not to compare the final laws by a transport inequality.

The second step is a stability estimate for infinitesimal perturbations of the interacting law. The appropriate quantity is the weighted negative Sobolev energy

$$\mathcal{E}_\rho(\sigma) = \sup_{\varphi \in C_c^\infty} \left\{ 2\sigma(\varphi) - \int |\nabla \varphi|^2 d\rho \right\}.$$

For a compatible family of marginal perturbations, let E_k denote the corresponding energy at level k . We derive a hierarchy of the form

$$E'_k \leq C E_k + q_k (E_{k+1} - E_k), \quad 0 \leq q_k \leq Ck, \quad (1.6)$$

with the last term absent at $k = N$. The difference $E_{k+1} - E_k$ is a squared orthogonal remainder. Keeping this difference, rather than bounding the two energies separately, allows the profile k^2/N^2 to be propagated by a direct comparison argument.

Finally, fix an observation time t and interpolate between the two evolutions by

$$\nu_s^N = S_{t-s}^{N,*} R_s^N, \quad 0 \leq s \leq t, \quad (1.7)$$

where S^N is the interacting semigroup. Its endpoints are $P_{N,t}$ and R_t^N . The derivative of this curve is the generator difference, propagated by S^N . Combining (1.5) and (1.6) bounds its k -particle energy by $C_T(1+s^{-1})k^2/N^2$. The square root of this bound is integrable at zero, which yields a Wasserstein length of order k/N . A synchronous coupling controls the remaining distance from $R_t^{N,(k)}$ to $\mu_t^{\otimes k}$.

Section 2 states the main theorem and gives the optimality example. Section 3 collects the tangent and entropy estimates. The generator difference is estimated in Section 4, and the tangent hierarchy is proved in Section 5. Section 6 completes the proof. The approximation needed to extend the tangent calculation to singular laws on Euclidean space is given in Appendix A.

2. MAIN RESULT AND AN OPTIMALITY EXAMPLE

2.1. Notation and assumptions. The space of probability measures with finite second moment on $\mathbb{R}^n$ is denoted by $\mathcal{P}_2(\mathbb{R}^n)$. For $\rho, \eta \in \mathcal{P}_2(\mathbb{R}^n)$, let

$$W_2^2(\rho, \eta) = \inf_{\pi \in \Pi(\rho, \eta)} \int_{\mathbb{R}^n \times \mathbb{R}^n} |x - y|^2 \pi(dx, dy), \quad (2.1)$$

where $\Pi(\rho, \eta)$ is the set of their couplings. All vector norms and transport costs are Euclidean, with no normalization by the number of particles. Matrix norms are operator norms unless the Hilbert–Schmidt norm is indicated.

Throughout the paper, $b \in C_b^\infty(\mathbb{R}^d \times \mathbb{R}^d; \mathbb{R}^d)$, meaning that b and all its derivatives are bounded. Fix nonnegative constants M, L_1, L_2 such that

$$\sup_{x,y} |b(x, y)| \leq M, \quad \sup_{x,y} \|D_1 b(x, y)\|_{\text{op}} \leq L_1, \quad \sup_{x,y} \|D_2 b(x, y)\|_{\text{op}} \leq L_2. \quad (2.2)$$

Coordinate sup-norm bounds $M_\infty, L_{1,\infty}, L_{2,\infty}$ imply (2.2) with $M = \sqrt{d} M_\infty$ and $L_i = \sqrt{d} L_{i,\infty}$. Thus the choice of norm changes only dimension-dependent factors in the constants. No bound on a derivative of order two or higher will enter the final estimate.

For $x = (x_1, \dots, x_N)$, write

$$b_N(x)_i = \frac{1}{N} \sum_{j=1}^N b(x_i, x_j), \quad L_N = \Delta + b_N \cdot \nabla, \quad L_0^N(s) = \Delta + \sum_{i=1}^N B_s(x_i) \cdot \nabla_i. \quad (2.3)$$

The diagonal term $j = i$ is included throughout. The diffusion coefficient $\sqrt{2}$ in (1.1) and (1.2) accounts for the coefficient one in front of Δ .

The equations are well posed under these assumptions for initial laws in $\mathcal{P}_2$; see [10]. The interacting semigroup is denoted by S_r^N , acting on functions, and its action on measures by $S_r^{N,*}$. The reference transition family $Q_{r,s}$ acts on measures on the right. In particular,

$$P_{N,t} = S_t^{N,*} P_{N,0}, \quad \mu_s = \mu_0 Q_{0,s}.$$

The law $P_{N,t}$ remains exchangeable whenever $P_{N,0}$ is exchangeable.

Theorem 2.1. *Let $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$ and, for each $N \geq 1$, let $P_{N,0} \in \mathcal{P}_2((\mathbb{R}^d)^N)$ be exchangeable. Suppose that, for some $C_0 \geq 0$,*

$$W_2^2(P_{N,0}^{(k)}, \mu_0^{\otimes k}) \leq C_0 \frac{k^2}{N^2}, \quad N \geq 1, \quad 1 \leq k \leq N. \quad (2.4)$$

Then, for every $T > 0$, there exists a finite constant C_T , depending only on d, C_0, T, M, L_1, L_2 , such that

$$\sup_{0 \leq t \leq T} W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq C_T \frac{k^2}{N^2}, \quad N \geq 1, \quad 1 \leq k \leq N. \quad (2.5)$$

The same constant works for all kernels and initial laws satisfying the stated bounds. An explicit choice is given in (6.11).

Remark 2.2. The assumption is made at every marginal level, including $k = N$. In the proof, information at levels $2k$ and N controls conditional entropy increments at level k . We do not assert that these all-level bounds are necessary for the conclusion, only that they are used by the argument. The theorem requires neither initial entropic chaos nor a transport or logarithmic Sobolev inequality for the reference law. It allows correlated or singular initial laws, but makes no claim of uniformity as $T \rightarrow \infty$.

Remark 2.3 (Weak Fokker–Planck formulations). The theorem also applies when the evolutions are specified as weak solutions rather than as SDE laws. More precisely, consider weakly continuous probability curves with locally uniformly bounded second moments satisfying

$$\langle \rho_t - \rho_0, \varphi \rangle = \int_0^t \langle \rho_s, \Delta \varphi + v_s \cdot \nabla \varphi \rangle ds, \quad \varphi \in C_c^\infty, \quad (2.6)$$

with $v = b_N$ or $v_s = B_s$. These curves coincide with the corresponding diffusion laws. To see this, first observe that weak continuity of μ_s and bounded continuity of b imply joint continuity of $B_s(x)$; the uniform spatial Lipschitz bound is L_1 . For a terminal compactly supported smooth test, its backward diffusion transition solves the backward equation and has bounded spatial derivatives on a finite interval. Testing (2.6) against spatial cutoffs of this backward solution, and then letting

the cutoff radius tend to infinity, identifies the terminal pairing with the initial pairing under the transition. Boundedness of the drift and of the test derivatives controls the cutoff errors. Time partitions justify the use of the time-dependent test. This proves the claimed identification. Compact-test continuity together with the stated moment condition would also suffice, since it implies weak continuity.

2.2. Optimality for correlated initial data. The following example shows that the order in (2.5) cannot be improved for the stated class of initial laws. It is already visible for independent heat evolutions.

Proposition 2.4. *Let $d = 1$, $b = 0$, and $\mu_0 = \mathcal{N}(0, 1)$. There are exchangeable initial laws satisfying (2.4) with $C_0 = 1/4$ such that, with $a_t = 1 + 2t$,*

$$W_2^2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) = \frac{(k/N)^2}{(\sqrt{a_t} + k/N + \sqrt{a_t})^2}, \quad t \geq 0. \quad (2.7)$$

Consequently, for each fixed $t \geq 0$, the right-hand side is bounded above and below by positive multiples of $(k/N)^2$, uniformly in N and $1 \leq k \leq N$.

Proof. Take independent standard Gaussian random variables $Z_0, Z_1, \dots, Z_N$, independent also of the Brownian motions, and set

$$X_i^N(0) = Z_i + N^{-1/2}Z_0.$$

At time t , the covariance matrices of the two laws in (2.7) are

$$a_t I_k + N^{-1} \mathbf{1} \mathbf{1}^T \quad \text{and} \quad a_t I_k,$$

respectively. Put $e = k^{-1/2} \mathbf{1}$ and $r = k/N$. For any coupling (X, Y) of these centered Gaussian laws, Cauchy–Schwarz gives

$$\mathbb{E}|X - Y|^2 \geq \mathbb{E}|e \cdot X - e \cdot Y|^2 \geq (\sqrt{a_t + r} - \sqrt{a_t})^2.$$

This lower bound is attained. Indeed, for a standard Gaussian vector $Z \in \mathbb{R}^k$, take

$$Y = \sqrt{a_t} Z, \quad X = \sqrt{a_t} Z + (\sqrt{a_t + r} - \sqrt{a_t})(e \cdot Z)e.$$

These random vectors have the required covariance matrices. Rationalizing the difference of square roots yields (2.7). At $t = 0$, its denominator is at least 4, proving the initial estimate. Since $0 < r \leq 1$, at a fixed time it lies between $4a_t$ and $(\sqrt{a_t + 1} + \sqrt{a_t})^2$. $\square$

3. TANGENT ENERGIES AND PRELIMINARY ESTIMATES

3.1. Weighted negative Sobolev norms. For a probability measure ρ on $\mathbb{R}^n$ and a linear functional σ on $C_c^\infty(\mathbb{R}^n)$, define

$$\mathcal{E}_\rho(\sigma) = \|\sigma\|_{-1,\rho}^2 := \sup_{\varphi \in C_c^\infty(\mathbb{R}^n)} \left\{ 2\sigma(\varphi) - \int |\nabla \varphi|^2 d\rho \right\}. \quad (3.1)$$

The definition does not require a density for ρ . Let H_ρ be the closure of compactly supported smooth gradients in $L^2(\rho; \mathbb{R}^n)$.

Lemma 3.1. *If $\mathcal{E}_\rho(\sigma) < \infty$, there is a unique $w \in H_\rho$ such that*

$$\sigma(\varphi) = \int \nabla \varphi \cdot w d\rho, \quad \|w\|_{L^2(\rho)}^2 = \mathcal{E}_\rho(\sigma). \quad (3.2)$$

If $V \in L^2(\rho)$ represents the same functional, then

$$\|V\|_{L^2(\rho)}^2 = \mathcal{E}_\rho(\sigma) + \|V - w\|_{L^2(\rho)}^2. \quad (3.3)$$

Moreover, $\mathcal{E}$ is lower semicontinuous when the base measures converge weakly and the source pairings converge on every compactly supported smooth test.

Proof. Replacing φ by a scalar multiple in (3.1) gives

$$|\sigma(\varphi)|^2 \leq \mathcal{E}_\rho(\sigma) \int |\nabla \varphi|^2 d\rho.$$

Thus $\nabla \varphi \mapsto \sigma(\varphi)$ extends to a continuous linear functional on H_ρ . Its Riesz representative is w . Completing the square and using the definition of H_ρ proves (3.2). Any other representative differs from w by a vector orthogonal to H_ρ , which gives (3.3). Finally, each fixed-test expression in (3.1) is continuous under the stated convergence. Taking the supremum proves lower semicontinuity. $\square$

Suppose now that ρ_N is a probability on $(\mathbb{R}^d)^N$ and $\sigma_N = -\operatorname{div}(\rho_N V)$, with $V \in L^2(\rho_N; (\mathbb{R}^d)^N)$. Its marginal source at level k is defined by

$$\sigma_k(\varphi) = \int \sum_{i=1}^k \nabla_i \varphi(x_1, \dots, x_k) \cdot V_i(x) \rho_N(dx). \quad (3.4)$$

Conditional expectation supplies an $L^2(\rho_k)$ representative. Let w_k be the minimal representative from Lemma 3.1, and put $E_k = \mathcal{E}_{\rho_k}(\sigma_k)$. Compatibility of the sources implies

$$\int w_{k+1} \cdot (w_k, 0) d\rho_{k+1} = E_k, \quad \|w_{k+1} - (w_k, 0)\|_{L^2(\rho_{k+1})}^2 = E_{k+1} - E_k. \quad (3.5)$$

To justify the first identity, approximate w_k by gradients in $L^2(\rho_k)$ and use (3.4); the second follows by expanding the square. In particular, $E_{k+1} \geq E_k$. No product structure is used here.

We will also use bounded smooth tests with bounded gradient, including cylindrical and periodic tests. Their pairings are defined by the representing field. Multiplying such a test by smooth spatial cutoffs approximates its gradient in $L^2(\rho)$, so the energy inequality extends to these tests. On a flat torus, the same definitions and identities hold with smooth periodic tests in place of compactly supported ones.

3.2. From tangent energy to transport. The next lemma is a convenient consequence of the dynamic characterization of Wasserstein distance [1, 2]. We include the reduction from the variational energy to a measurable velocity field.

Lemma 3.2. *Let $\rho : [0, t] \rightarrow \mathcal{P}_2(\mathbb{R}^n)$ be W_2 -continuous. Suppose that linear sources σ_s have locally integrable compact-test pairings on $(0, t)$ and satisfy*

$$\langle \rho_v - \rho_u, \varphi \rangle = \int_u^v \sigma_s(\varphi) ds, \quad 0 < u < v \leq t, \quad \varphi \in C_c^\infty(\mathbb{R}^n). \quad (3.6)$$

If $g : (0, t] \rightarrow [0, \infty)$ is continuous, $\int_0^t g(s) ds < \infty$, and

$$\mathcal{E}_{\rho_s}(\sigma_s) \leq g(s)^2 \quad \text{for a.e. } s \in (0, t),$$

then

$$W_2(\rho_0, \rho_t) \leq \int_0^t g(s) ds. \quad (3.7)$$

Proof. First suppose the energy is bounded by a constant A on a compact interval $[u, v] \subset (0, t]$. With $dM(s, x) = ds \rho_s(dx)$, define, on finite sums of time-weighted test gradients,

$$\mathcal{F}(\nabla \varphi) = \int_u^v \sigma_s(\varphi_s) ds.$$

Lemma 3.1 and Cauchy–Schwarz show that

$$|\mathcal{F}(\nabla \varphi)| \leq \sqrt{A(v-u)} \|\nabla \varphi\|_{L^2(M)}.$$

The Riesz representation theorem in $L^2(M)$ therefore gives a jointly measurable field U representing $\mathcal{F}$, with

$$\int_u^v \int |U_s|^2 d\rho_s ds \leq A(v-u).$$

Using time indicator functions identifies (3.6) with the distributional equation $\partial_s \rho_s + \operatorname{div}(\rho_s U_s) = 0$. The dynamic transport inequality [1, Theorem 8.3.1], followed by Cauchy–Schwarz, yields

$$W_2(\rho_u, \rho_v) \leq \int_u^v \|U_s\|_{L^2(\rho_s)} ds \leq (v - u) \sqrt{A}. \quad (3.8)$$

Now partition $[\varepsilon, t]$ and apply (3.8) on each interval with A equal to the supremum of g^2 there. The triangle inequality and convergence of upper Riemann sums give

$$W_2(\rho_\varepsilon, \rho_t) \leq \int_\varepsilon^t g(s) ds.$$

Let $\varepsilon \downarrow 0$ and use W_2 -continuity at zero. Notice that only g , not g^2 , is required to be integrable at zero. $\square$

3.3. Entropy–cost regularization. For probability measures R, S on the same space, write

$$H(R | S) = \int \log\left(\frac{dR}{dS}\right) dR$$

when $R \ll S$, and set it equal to $+\infty$ otherwise.

Lemma 3.3. *Let Q be the transition family of*

$$dZ(r) = a_r(Z(r)) dr + \sqrt{2} dW(r),$$

where a is bounded, jointly continuous, and uniformly L -Lipschitz in space. For $s > 0$, $m \geq 1$, and $R, S \in \mathcal{P}_2((\mathbb{R}^d)^m)$,

$$H(RQ_{0,s}^{\otimes m} | SQ_{0,s}^{\otimes m}) \leq \beta_L(s) W_2^2(R, S), \quad \beta_L(s) = \frac{1 + Ls + L^2 s^2 / 3}{4s}. \quad (3.9)$$

The coefficient is independent of m .

Proof. The block drift on $(\mathbb{R}^d)^m$ has the same Lipschitz constant L , so it suffices to prove the corresponding estimate in any finite dimension. Fix starting points x, y , let $D = y - x$, and put $\delta_r = (1 - r/s)D$. Compare two path laws, both started at x , with respective drifts

$$a_r(z) \quad \text{and} \quad \tilde{a}_r(z) = a_r(z + \delta_r) + D/s.$$

Under the second law, $Z(r) + \delta_r$ is the ordinary diffusion started at y , and its terminal value is $Z(s)$. Denote these path laws by $\mathbb{P}_x$ and $\tilde{\mathbb{P}}_{x,y}$. For fixed x, y , the drift difference is bounded, so Girsanov's theorem gives

$$\begin{aligned} H(\mathbb{P}_x | \tilde{\mathbb{P}}_{x,y}) &= \frac{1}{4} \mathbb{E}_{\mathbb{P}_x} \int_0^s |a_r(Z(r)) - a_r(Z(r) + \delta_r) - D/s|^2 dr \\ &\leq \frac{|D|^2}{4} \int_0^s [L(1 - r/s) + s^{-1}]^2 dr = \beta_L(s) |x - y|^2. \end{aligned} \quad (3.10)$$

The factor $1/4$ corresponds to the diffusion coefficient $\sqrt{2}$. Applying data processing to the terminal coordinate gives the same bound for $H(Q_{0,s}(x, \cdot) | Q_{0,s}(y, \cdot))$. Finally, integrate over any coupling of R and S and use joint convexity of relative entropy. Taking the infimum over couplings proves (3.9). The change of measure was used only for fixed starting points; no exponential moment of the initial coupling is needed. $\square$

3.4. Entropy increments under exchangeability.

Lemma 3.4. *Let R_N be exchangeable, let μ be a probability measure, and suppose $h_N < \infty$, where*

$$h_m = H(R_N^{(m)} | \mu^{\otimes m}), \quad h_0 = 0.$$

Then $\delta_m = h_m - h_{m-1}$ is nonnegative and nondecreasing. Consequently,

$$\delta_{m+1} \leq \frac{h_{2m}}{m} \quad (2m \leq N), \quad \delta_{m+1} \leq \frac{h_N}{N - m} \quad (m < N). \quad (3.11)$$

If $h_j \leq H j^2 / N^2$ for all j , then

$$\left(\frac{N-m}{N}\right)^2 m (h_{m+1} - h_m) \leq 4H \frac{m^2}{N^2}, \quad m < N. \quad (3.12)$$

Proof. For $X \sim R_N$, the entropy chain rule gives

$$\delta_{m+1} = \mathbb{E}H(\text{Law}(X_{m+1} \mid X_1, \dots, X_m) \mid \mu) \geq 0.$$

Convexity of entropy shows that removing X_m from the conditioning can only decrease this average. By exchangeability, the resulting expression equals δ_m . This proves monotonicity. Summing $\delta_j \geq \delta_{m+1}$ over $m < j \leq 2m$, or over $m < j \leq N$, and using $h_m \geq 0$, gives (3.11).

If $2m \leq N$, the first bound gives $m\delta_{m+1} \leq 4Hm^2/N^2$. If $2m > N$, the second gives

$$\left(\frac{N-m}{N}\right)^2 m\delta_{m+1} \leq H \frac{m(N-m)}{N^2} \leq H \frac{m^2}{N^2}.$$

Together these prove (3.12). $\square$

4. THE REGULARIZED GENERATOR DIFFERENCE

Fix $N \geq 1$ and $s > 0$. Recall the reference law

$$R_s^N = P_{N,0} Q_{0,s}^{\otimes N},$$

and abbreviate its m -particle marginal by $R_s^{(m)}$. Lemma 3.3, applied to the drift B with Lipschitz constant L_1 , gives

$$h_m(s) := H(R_s^{(m)} \mid \mu_s^{\otimes m}) \leq H_s \frac{m^2}{N^2}, \quad H_s := C_0 \beta_{L_1}(s). \quad (4.1)$$

In particular, all these entropies are finite, even when the corresponding initial entropies are infinite.

Define the centered interaction

$$c_s(x, y) = b(x, y) - B_s(x), \quad \int c_s(x, y) \mu_s(dy) = 0.$$

The difference between the reference and interacting generators is the source

$$\zeta_s = (L_0^N(s)^* - L_N^*) R_s^N = -\text{div}(R_s^N U_s), \quad (U_s)_i(x) = -\frac{1}{N} \sum_{j=1}^N c_s(x_i, x_j). \quad (4.2)$$

The diffusion terms cancel. By exchangeability, an $L^2(R_s^{(m)})$ representative of the m -particle marginal source is

$$(g_s^{(m)})_i(x) = -\frac{1}{N} \sum_{j=1}^m c_s(x_i, x_j) - \alpha_m \mathbb{E}_{R_s^N} [c_s(x_i, X_{m+1}) \mid X_{1:m} = x], \quad \alpha_m = \frac{N-m}{N}, \quad (4.3)$$

for $i \leq m < N$. The second term is absent when $m = N$.

Proposition 4.1. *Under the assumptions of Theorem 2.1,*

$$\mathcal{E}_{R_s^{(m)}}(\zeta_s^{(m)}) \leq A_b(1 + H_s) \frac{m^2}{N^2}, \quad A_b := 48dM^2, \quad 1 \leq m \leq N. \quad (4.4)$$

Proof. If $M = 0$, the source vanishes and the result is immediate. Assume $M > 0$. By Lemma 3.1, it is enough to estimate the squared norm of (4.3). We treat the internal sum and the conditional term separately.

The conditional term. Let $m < N$ and let λ_x be the conditional law of X_{m+1} given $X_{1:m} = x$ under R_s^N . We use the convention $\|\lambda - \mu\|_{\text{TV}} = \sup_A |\lambda(A) - \mu(A)|$. Since $|b| \leq M$, Pinsker's inequality gives

$$\left| \int b(x_i, y) (\lambda_x - \mu_s)(dy) \right|^2 \leq 2M^2 H(\lambda_x \mid \mu_s). \quad (4.5)$$

For the vector-valued integral, this follows by testing against a unit vector and using the scalar bound. Summing over i , integrating over $R_s^{(m)}$, and applying the entropy chain rule yields

$$\int \sum_{i=1}^m |\mathbb{E}[c_s(x_i, X_{m+1}) \mid X_{1:m} = x]|^2 R_s^{(m)}(dx) \leq 2M^2 m [h_{m+1}(s) - h_m(s)].$$

Thus (3.12) implies

$$X_m := \alpha_m^2 \int \sum_{i=1}^m |\mathbb{E}[c_s(x_i, X_{m+1}) \mid X_{1:m}]|^2 dR_s^{(m)} \leq 8M^2 H_s \frac{m^2}{N^2}. \quad (4.6)$$

Set $X_N = 0$.

The internal sum. For $i \leq m$, put $S_i = \sum_{j=1}^m c_s(x_i, x_j)$. Under $\mu_s^{\otimes m}$, conditional on x_i , the scalar coordinates of the terms with $j \neq i$ are independent and centered. Each such scalar term lies in an interval of length at most $2M$. If $S_{i,a}^\circ$ denotes their sum in coordinate a , Hoeffding's exponential estimate [4] gives

$$\mathbb{E}_{\mu_s^{\otimes m}}[e^{u S_{i,a}^\circ} \mid x_i] \leq \exp\left(\frac{(m-1)M^2 u^2}{2}\right). \quad (4.7)$$

The remaining diagonal term has absolute value at most $2M$. For $\lambda = (16mM^2)^{-1}$, the inequality $(v+w)^2 \leq 2v^2 + 2w^2$ and the Gaussian identity $e^{\eta z^2} = \mathbb{E}e^{\sqrt{2\eta} z G}$, with $G \sim \mathcal{N}(0, 1)$, give

$$\begin{aligned} \mathbb{E}[e^{\lambda S_{i,a}^2} \mid x_i] &\leq e^{8\lambda M^2} (1 - 4\lambda(m-1)M^2)^{-1/2} \\ &\leq e^{1/2} \sqrt{2} \leq e. \end{aligned} \quad (4.8)$$

This also covers $m = 1$. Jensen's inequality over the d coordinates now shows that

$$\int \exp\left(\frac{|S_i|^2}{16dmM^2}\right) d\mu_s^{\otimes m} \leq e. \quad (4.9)$$

Applying the entropy variational inequality to this exponential moment gives

$$\int |S_i|^2 dR_s^{(m)} \leq 16dmM^2 [h_m(s) + 1].$$

Since $m \leq N$, (4.1) implies $h_m(s) \leq H_s$. Summing over i and dividing by N^2 , we obtain

$$I_m := \int \sum_{i=1}^m \left| \frac{S_i}{N} \right|^2 dR_s^{(m)} \leq 16dM^2(1 + H_s) \frac{m^2}{N^2}. \quad (4.10)$$

Combining (4.6) and (4.10),

$$\mathcal{E}_{R_s^{(m)}}(\zeta_s^{(m)}) \leq \|g_s^{(m)}\|_{L^2(R_s^{(m)})}^2 \leq 2I_m + 2X_m \leq 48dM^2(1 + H_s) \frac{m^2}{N^2}.$$

The same bound holds at $m = N$, where only the internal term remains. $\square$

5. PROPAGATION OF TANGENT ENERGIES

The estimate in this section concerns a general perturbation of an interacting law. It will be applied to the source in Proposition 4.1.

Let $\rho_{N,0} \in \mathcal{P}_2((\mathbb{R}^d)^N)$ be exchangeable and let $V \in L^2(\rho_{N,0}; (\mathbb{R}^d)^N)$ be permutation-equivariant, up to a $\rho_{N,0}$ -null set. Let $F_r(x, W)$ be the stochastic flow of (1.1), and write $J_r(x, W) = D_x F_r(x, W)$. For $X_0 \sim \rho_{N,0}$ independent of W , define

$$\begin{aligned} \rho_{N,r} &= \text{Law}(F_r(X_0, W)), \\ \sigma_{N,r}(\varphi) &= \mathbb{E}[\nabla \varphi(F_r(X_0, W)) \cdot J_r(X_0, W) V(X_0)]. \end{aligned} \quad (5.1)$$

Equivalently, $\sigma_{N,r}(\varphi) = \sigma_{N,0}(S_r^N \varphi)$, where $\sigma_{N,0} = -\text{div}(\rho_{N,0} V)$. Both $\rho_{N,r}$ and $\sigma_{N,r}$ therefore evolve under the same linear generator L_N^* . Their marginal families are compatible and exchangeable.

Set

$$K = L_1 + L_2, \quad G = 2L_1^2 + L_2^2 + 2M^2, \quad C = \max\{2K + 2L_1 + 1, G\}. \quad (5.2)$$

The derivative of the interacting drift satisfies $\|Db_N\|_{\text{op}} \leq K$. Indeed,

$$(Db_N(x)v)_i = \frac{1}{N} \sum_{j=1}^N D_1 b(x_i, x_j) v_i + \frac{1}{N} \sum_{j=1}^N D_2 b(x_i, x_j) v_j,$$

and the two terms have Euclidean operator norm at most L_1 and L_2 , respectively. Hence $\|J_r\|_{\text{op}} \leq e^{Kr}$. Conditional expectation in (5.1) supplies L^2 representatives at every level. In particular,

$$E_m(r) := \mathcal{E}_{\rho_{N,r}^{(m)}}(\sigma_{N,r}^{(m)}) \leq e^{2Kr} \int |V|^2 d\rho_{N,0}. \quad (5.3)$$

This bound ensures finiteness; the following proposition preserves the sharper marginal profile.

Proposition 5.1. *If, for some $A \geq 0$,*

$$E_m(0) \leq A \frac{m^2}{N^2}, \quad 1 \leq m \leq N, \quad (5.4)$$

then

$$E_m(r) \leq A e^{4Cr} \frac{m^2}{N^2}, \quad r \geq 0, \quad 1 \leq m \leq N, \quad (5.5)$$

where C is defined in (5.2).

We first derive the hierarchy for smooth positive laws on a flat torus. In this setting the minimal vector fields are gradients of solutions of weighted Poisson equations, and the energy can be differentiated directly. Appendix A extends the resulting estimate to the laws and sources in (5.1).

5.1. The smooth periodic calculation. Fix a period $P > 0$ and let $\mathbb{T}_P^d = \mathbb{R}^d / (P\mathbb{Z})^d$. Suppose b is separately P -periodic in its two variables. For this subsection, assume that $\rho_{N,0}$ has a smooth strictly positive density on $(\mathbb{T}_P^d)^N$, and that $\sigma_{N,0}$ has a smooth signed density with integral zero. Assume also that both are invariant under particle permutations. Evolve them by L_N^* . Their densities remain smooth, and $\rho_{N,r}$ remains strictly positive on every compact time interval.

Write ρ_m, σ_m for the marginals at time r . The weighted Poisson equation

$$-\text{div}(\rho_m \nabla f_m) = \sigma_m, \quad \int_{(\mathbb{T}_P^d)^m} f_m dx = 0, \quad (5.6)$$

has a unique smooth solution. Positivity of ρ_m gives coercivity on mean-zero functions, and elliptic regularity gives smoothness. Difference quotients in r satisfy a coercive elliptic equation on each compact time interval; passing to the limit gives differentiability of f_m in time. Thus

$$w_m = \nabla f_m, \quad E_m = \int |\nabla f_m|^2 d\rho_m.$$

The constants used to solve (5.6) need not be uniform in P, N , or the lower bound on ρ_m ; none will occur in the following estimate.

Lemma 5.2. *In the smooth periodic setting above, put*

$$q_m = Gm \frac{N-m}{N}, \quad 1 \leq m < N, \quad q_N = 0.$$

Then

$$E'_m \leq CE_m + q_m(E_{m+1} - E_m) \quad (m < N), \quad E'_N \leq CE_N. \quad (5.7)$$

The constants are independent of the period and of lower bounds on the densities.

Proof. Fix m and abbreviate $f = f_m$ at the time under consideration. Define the internal drift and generator by

$$B_m(x)_i = \frac{1}{N} \sum_{j=1}^m b(x_i, x_j), \quad \mathcal{L}_m = \Delta + B_m \cdot \nabla,$$

and, for a smooth m -particle function ψ , set

$$\Phi_\psi(x, y) = \sum_{i=1}^m b(x_i, y) \cdot \nabla_i \psi(x).$$

Here $x = (x_1, \dots, x_m)$ and y is the next particle coordinate. Exchangeability gives the two marginal equations

$$\begin{aligned} \frac{d}{dr} \rho_m(\psi) &= \rho_m(\mathcal{L}_m \psi) + \alpha_m \rho_{m+1}(\Phi_\psi), \\ \frac{d}{dr} \sigma_m(\psi) &= \sigma_m(\mathcal{L}_m \psi) + \alpha_m \sigma_{m+1}(\Phi_\psi), \quad \alpha_m = \frac{N-m}{N}. \end{aligned} \quad (5.8)$$

The second equation is available because the perturbation is propagated by the same full generator as the base law. Both external terms are absent at $m = N$.

Differentiating $E_m = \sigma_m(f) = \rho_m(|\nabla f|^2)$ and using (5.6) to cancel the terms containing $\partial_r f$ gives

$$E'_m = 2\sigma'_m(f) - \rho'_m(|\nabla f|^2). \quad (5.9)$$

Substituting (5.8) into (5.9),

$$\begin{aligned} E'_m &= 2 \int \nabla f \cdot \nabla \mathcal{L}_m f \, d\rho_m - \int \mathcal{L}_m |\nabla f|^2 \, d\rho_m \\ &\quad + \alpha_m \left[2 \int w_{m+1} \cdot \nabla_{x,y} \Phi_f \, d\rho_{m+1} - \int \sum_{i=1}^m b(x_i, y) \cdot \nabla_i |\nabla f|^2 \, d\rho_{m+1} \right]. \end{aligned} \quad (5.10)$$

All generator actions in this identity are applied with f held fixed.

Diffusion and internal drift. Let $\mathcal{H}_m = \int |D^2 f|_{\text{HS}}^2 \, d\rho_m$. The pointwise Bochner identity shows that the first line of (5.10) is

$$-2\mathcal{H}_m + 2 \int \nabla f \cdot (DB_m)^\top \nabla f \, d\rho_m \leq -2\mathcal{H}_m + 2KE_m.$$

For the last inequality, note that $\|DB_m\|_{\text{op}} \leq (m/N)(L_1 + L_2) \leq K$. This calculation does not involve derivatives of the density.

External interaction. Decompose

$$w_{m+1} = (\nabla f, 0) + \mathcal{R}_m.$$

The orthogonal increment identity (3.5) gives

$$\|\mathcal{R}_m\|_{L^2(\rho_{m+1})}^2 = E_{m+1} - E_m. \quad (5.11)$$

The contribution of $(\nabla f, 0)$ cancels the Hessian terms in the external bracket:

$$\begin{aligned} &2(\nabla f, 0) \cdot \nabla_{x,y} \Phi_f - \sum_{i=1}^m b(x_i, y) \cdot \nabla_i |\nabla f|^2 \\ &= 2 \sum_{i=1}^m \nabla_i f \cdot D_1 b(x_i, y)^\top \nabla_i f \leq 2L_1 |\nabla f|^2. \end{aligned} \quad (5.12)$$

To estimate the remaining term, differentiate Φ_f . In the x variables, the derivative consists of a block diagonal $D_1 b$ term and $D^2 f$ applied to the vector $(b(x_i, y))_{i \leq m}$. The latter vector has norm

at most $\sqrt{m}M$. In the y variable, it is $\sum_i D_2 b(x_i, y)^\top \nabla_i f$. Hence

$$\begin{aligned} |\nabla_x \Phi_f|^2 &\leq 2L_1^2 |\nabla f|^2 + 2mM^2 |D^2 f|_{\text{HS}}^2, \\ |\nabla_y \Phi_f|^2 &\leq mL_2^2 |\nabla f|^2, \\ |\nabla_{x,y} \Phi_f|^2 &\leq mG(|\nabla f|^2 + |D^2 f|_{\text{HS}}^2). \end{aligned} \quad (5.13)$$

By Cauchy–Schwarz, (5.11), and $2ab \leq a^2 + b^2$,

$$\begin{aligned} 2 \left| \int \mathcal{R}_m \cdot \nabla_{x,y} \Phi_f d\rho_{m+1} \right| &\leq 2\sqrt{mG(E_{m+1} - E_m)} \sqrt{E_m + \mathcal{H}_m} \\ &\leq mG(E_{m+1} - E_m) + E_m + \mathcal{H}_m. \end{aligned} \quad (5.14)$$

Combining these bounds in (5.10) gives

$$\begin{aligned} E'_m &\leq (2K + 2\alpha_m L_1 + \alpha_m) E_m + \alpha_m mG(E_{m+1} - E_m) - (2 - \alpha_m) \mathcal{H}_m \\ &\leq C E_m + q_m (E_{m+1} - E_m). \end{aligned}$$

At $m = N$, only the diffusion and internal drift terms occur, and $E'_N \leq 2K E_N \leq C E_N$. $\square$

The use of the full next-level gradient in (5.11) is essential to this calculation. It yields the difference $E_{m+1} - E_m$ before any estimate of the external interaction. Bounding w_{m+1} on its own would not give the same closed comparison for a quadratic marginal profile.

5.2. Comparison of the hierarchy.

Proof of Proposition 5.1. First work in the smooth periodic setting of Lemma 5.2. An integrating factor gives, for $m < N$,

$$E_m(r) \leq e^{(C-q_m)r} E_m(0) + q_m \int_0^r e^{(C-q_m)(r-u)} E_{m+1}(u) du, \quad (5.15)$$

and $E_N(r) \leq e^{Cr} E_N(0)$. Consider the comparison functions

$$\bar{E}_m(r) = A e^{4Cr} \frac{m^2}{N^2}.$$

Since $q_m \leq Cm$ and $2m + 1 \leq 3m$ for $m \geq 1$,

$$q_m((m+1)^2 - m^2) \leq 3Cm^2.$$

It follows that

$$\bar{E}'_m \geq (C - q_m) \bar{E}_m + q_m \bar{E}_{m+1}, \quad \bar{E}_m(0) \geq E_m(0).$$

The comparison at $m = N$ is immediate. Backward induction in (5.15) then yields $E_m \leq \bar{E}_m$ for every m . The integral kernel is nonnegative, regardless of the sign of $C - q_m$.

To remove the smoothness and periodicity restrictions, Appendix A gives three approximation steps. Product heat convolution on a fixed torus regularizes the base and the source without increasing any initial marginal energy. Lower semicontinuity then extends the estimate to arbitrary initial torus laws. Increasing the test period recovers the full Euclidean energy for periodic kernels. Finally, a periodic approximation of b preserving (2.2), together with convergence of the stochastic flows and their first variations, gives the result for the original kernel. These steps preserve the constant in (5.5) and apply to the source defined in (5.1). $\square$

6. PROOF OF THE MAIN THEOREM

Fix $0 < t \leq T$ and consider the curve

$$\nu_s^N = S_{t-s}^{N,*} R_s^N, \quad R_s^N = P_{N,0} Q_{0,s}^{\otimes N}, \quad 0 \leq s \leq t. \quad (6.1)$$

Its endpoints are

$$\nu_0^N = P_{N,t}, \quad \nu_t^N = R_t^N. \quad (6.2)$$

We estimate the distance between these endpoints and then compare $R_t^{N,(k)}$ with $\mu_t^{\otimes k}$.

6.1. Derivative and tangent estimate. For a compactly supported smooth test φ on $(\mathbb{R}^d)^N$, differentiation of the two semigroup parameters gives

$$\begin{aligned} \frac{d}{ds} \langle \nu_s^N, \varphi \rangle &= \langle R_s^N, (L_0^N(s) - L_N) S_{t-s}^N \varphi \rangle \\ &= \int U_s \cdot \nabla S_{t-s}^N \varphi dR_s^N =: \tau_s^N(\varphi), \quad 0 < s < t. \end{aligned} \quad (6.3)$$

The backward transition and its required derivatives are bounded on a finite interval. Spatial cutoffs therefore justify the use of the weak forward equation in this identity. The same reasoning applies to bounded smooth cylindrical lifts of marginal tests. In particular, for $0 < a < b \leq t$,

$$\langle \nu_b^{N,(k)} - \nu_a^{N,(k)}, \varphi \rangle = \int_a^b \tau_s^{N,(k)}(\varphi) ds. \quad (6.4)$$

The source τ_s^N is precisely the source ζ_s from (4.2), propagated for time $t - s$ by the interacting dynamics. Indeed, if $X_s \sim R_s^N$ is independent of an interacting Brownian path W , then

$$\tau_s^N(\varphi) = \mathbb{E}[\nabla \varphi(F_{t-s}(X_s, W)) \cdot J_{t-s}(X_s, W) U_s(X_s)]. \quad (6.5)$$

For each fixed s , the field U_s is bounded and permutation-equivariant. Proposition 4.1 supplies its initial energy bound at every level, and Proposition 5.1 then gives

$$\mathcal{E}_{\nu_s^{N,(k)}}(\tau_s^{N,(k)}) \leq A_b e^{4C(t-s)} (1 + C_0 \beta_{L_1}(s)) \frac{k^2}{N^2}, \quad 0 < s \leq t. \quad (6.6)$$

The consistency needed for the tangent hierarchy comes from this single full source.

6.2. Continuity at the endpoints. To apply Lemma 3.2, we verify W_2 -continuity of the interpolation, including at $s = 0$. On one probability space take $X_0 \sim P_{N,0}$ and two independent N -particle Brownian paths $W^{(1)}, W^{(2)}$, independent of X_0 . Let $\bar{F}_s$ be the reference flow with drift B , acting coordinatewise, and define

$$X_s = F_{t-s}(\bar{F}_s(X_0, W^{(1)}), W^{(2)}), \quad 0 \leq s \leq t. \quad (6.7)$$

Then $\text{Law}(X_s) = \nu_s^N$ for every s .

Boundedness of the drifts and the second-moment estimate for Brownian increments imply mean-square continuity of either flow in its time parameter. The interacting flow is also e^{KT} -Lipschitz in its starting point. Splitting the difference in (6.7) into a change of starting point and a change of elapsed interacting time therefore gives

$$\mathbb{E}|X_s - X_r|^2 \leq C_{N,d,M,K,T}(|s - r| + |s - r|^2), \quad r, s \in [0, t],$$

for a finite constant. All the laws have finite second moments. Thus $s \mapsto \nu_s^N$ is W_2 -continuous, as is each marginal curve. The possible dependence on N in this continuity estimate is harmless: it is used only to pass to an endpoint with N fixed, not to obtain the propagation-of-chaos rate.

6.3. The transport estimate. Define

$$D_T = \frac{1 + L_1 T + L_1^2 T^2 / 3}{4}. \quad (6.8)$$

For $0 < s \leq T$, one has $\beta_{L_1}(s) \leq D_T/s$. The square root of the right-hand side of (6.6) is therefore bounded by

$$g_k(s) = \sqrt{A_b} e^{2CT} \left(1 + \sqrt{\frac{C_0 D_T}{s}} \right) \frac{k}{N}.$$

This function is integrable at zero. Lemma 3.2, together with (6.4) and the endpoint continuity just proved, gives

$$W_2(P_{N,t}^{(k)}, R_t^{N,(k)}) \leq \sqrt{A_b} e^{2CT} (t + 2\sqrt{C_0 D_T t}) \frac{k}{N}. \quad (6.9)$$

The singularity in the energy costs only a square-root term in this length estimate.

It remains to compare the reference endpoint with the product law. Couple $P_{N,0}^{(k)}$ and $\mu_0^{\otimes k}$ arbitrarily and evolve both configurations with the same k reference Brownian motions. Since B_s is L_1 -Lipschitz, the noise cancels and Gronwall's inequality gives

$$\sum_{i=1}^k |\bar{X}_i(t) - \bar{Y}_i(t)|^2 \leq e^{2L_1 t} \sum_{i=1}^k |\bar{X}_i(0) - \bar{Y}_i(0)|^2.$$

Taking the infimum over initial couplings and using (2.4),

$$W_2(R_t^{N,(k)}, \mu_t^{\otimes k}) \leq e^{L_1 t} \sqrt{C_0} \frac{k}{N}. \quad (6.10)$$

Combining (6.9) and (6.10) by the triangle inequality proves (2.5). More explicitly, an admissible choice is

$$C_T = \left[e^{L_1 T} \sqrt{C_0} + \sqrt{48d} M e^{2CT} (T + 2\sqrt{C_0 D_T T}) \right]^2, \quad (6.11)$$

with C and D_T given in (5.2) and (6.8). This constant depends only on the quantities specified in Theorem 2.1. At $t = 0$, the initial estimate applies and $C_T \geq C_0$. This completes the proof. $\square$

Remark 6.1. For product initial data, $C_0 = 0$, the entropy regularization has no singular contribution. The proof then gives the more informative short-time estimate

$$W_2(P_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq \sqrt{48d} M e^{2Ct} t \frac{k}{N}.$$

No optimality in the time dependence or in the numerical constant is asserted.

APPENDIX A. APPROXIMATION IN THE TANGENT ESTIMATE

This appendix completes the extension used in Proposition 5.1. The point is to regularize the full base and source together while preserving the initial energy bound at every marginal level. The differential calculation is then needed only for smooth positive densities.

A.1. Arbitrary initial laws on a fixed torus. Work first on $(\mathbb{T}_P^d)^N$, with a smooth periodic kernel. Let $\rho_{N,0}$ be any exchangeable probability measure and let $\sigma_{N,0}$ be an exchangeable source with an $L^2(\rho_{N,0})$ representative. Let $\mathcal{K}_\varepsilon^{(m)}$ denote heat convolution at time $\varepsilon > 0$ on $(\mathbb{T}_P^d)^m$. At the full level, define

$$\rho_{N,0}^\varepsilon = \mathcal{K}_\varepsilon^{(N)} \rho_{N,0}, \quad \sigma_{N,0}^\varepsilon(\psi) = \sigma_{N,0}(\mathcal{K}_\varepsilon^{(N)} \psi). \quad (A.1)$$

The base has a smooth strictly positive density because the torus heat kernel is strictly positive. The source has a smooth signed density of total mass zero. Product convolution preserves exchangeability and commutes with taking marginals, so

$$\rho_{m,0}^\varepsilon = \mathcal{K}_\varepsilon^{(m)} \rho_{m,0}, \quad \sigma_{m,0}^\varepsilon = \mathcal{K}_\varepsilon^{(m)} \sigma_{m,0}.$$

For a smooth periodic m -particle test φ , symmetry of heat convolution and Jensen's inequality give

$$\begin{aligned} 2\sigma_{m,0}^\varepsilon(\varphi) - \int |\nabla \varphi|^2 d\rho_{m,0}^\varepsilon &= 2\sigma_{m,0}(\mathcal{K}_\varepsilon^{(m)} \varphi) - \int \mathcal{K}_\varepsilon^{(m)}(|\nabla \varphi|^2) d\rho_{m,0} \\ &\leq 2\sigma_{m,0}(\mathcal{K}_\varepsilon^{(m)} \varphi) - \int |\nabla \mathcal{K}_\varepsilon^{(m)} \varphi|^2 d\rho_{m,0} \\ &\leq \mathcal{E}_{\rho_{m,0}}(\sigma_{m,0}). \end{aligned} \quad (A.2)$$

Thus

$$\mathcal{E}_{\rho_{m,0}^\varepsilon}(\sigma_{m,0}^\varepsilon) \leq \mathcal{E}_{\rho_{m,0}}(\sigma_{m,0}) \leq A \frac{m^2}{N^2}, \quad 1 \leq m \leq N. \quad (A.3)$$

This is an all-level contraction, not merely a bound on the full energy.

Evolve both regularized objects with the same interacting semigroup. They satisfy the smooth periodic hypotheses, so the comparison in Section 5 gives

$$\mathcal{E}_{\rho_{m,r}^\varepsilon}(\sigma_{m,r}^\varepsilon) \leq Ae^{4Cr} \frac{m^2}{N^2}.$$

For each fixed r and each smooth full test ψ ,

$$\sigma_{N,r}^\varepsilon(\psi) = \sigma_{N,0}(\mathcal{K}_\varepsilon^{(N)} S_r^N \psi) \longrightarrow \sigma_{N,0}(S_r^N \psi) = \sigma_{N,r}(\psi).$$

Indeed, $S_r^N \psi$ is smooth and periodic, heat convolution converges in C^1 , and $\sigma_{N,0}$ is represented by an integrable field. The base measures converge weakly as well. Applying the torus version of Lemma 3.1 to each marginal proves (5.5) for arbitrary initial torus laws.

A.2. Recovery of the Euclidean energy for periodic kernels. Suppose b is separately P -periodic, but the initial law and the stochastic flows are on Euclidean space as in (5.1). Project the law modulo P and define the projected source by its pairings with periodic tests. The periodic drift ensures that projection commutes with the interacting semigroup. Consequently, these projected objects are exactly the torus evolution considered above.

Let $E_m^P(r)$ be the energy obtained by taking the supremum over smooth P -periodic tests, paired against the original Euclidean law and source. The cutoff observation following (3.5) shows that $E_m^P(0) \leq E_m(0)$. The torus estimate therefore yields

$$E_m^P(r) \leq Ae^{4Cr} \frac{m^2}{N^2}. \quad (\text{A.4})$$

A fixed periodic test class need not recover the full Euclidean energy. To do so, note that the same kernel is periodic with every period $P_j = jP$, $j \geq 1$. The constant in (A.4) is independent of the period.

Fix $\varphi \in C_c^\infty((\mathbb{R}^d)^m)$. For all sufficiently large j , its support lies strictly inside the centered fundamental cube of side P_j . Let φ_j be its P_j -periodic extension. Since φ vanishes near the boundary of the cube, this extension is smooth, and

$$\nabla \varphi_j(x) \longrightarrow \nabla \varphi(x), \quad |\nabla \varphi_j(x)| \leq \|\nabla \varphi\|_\infty.$$

The source at time r has an L^2 representative by (5.3). Dominated convergence therefore gives

$$2\sigma_{m,r}(\varphi) - \int |\nabla \varphi|^2 d\rho_{m,r} = \lim_{j \rightarrow \infty} \left[2\sigma_{m,r}(\varphi_j) - \int |\nabla \varphi_j|^2 d\rho_{m,r} \right] \leq Ae^{4Cr} \frac{m^2}{N^2}.$$

Taking the supremum over compactly supported tests proves the desired bound for the full Euclidean energy. In particular, the resulting estimate concerns Euclidean laws and Euclidean transport costs, not their torus projections.

A.3. Approximation of a general kernel. For $R \geq 1$, let $\theta_R(x) = (R \sin(x_a/R))_{a=1}^d$ and define the periodic kernel

$$b_R(x, y) = b(\theta_R(x), \theta_R(y)), \quad R \geq 1. \quad (\text{A.5})$$

These kernels are smooth and separately $2\pi R$ -periodic. Since $\|D\theta_R\|_{\text{op}} \leq 1$, they satisfy the same bounds M, L_1, L_2 as b . Moreover, $b_R \rightarrow b$ in C^1 on compact subsets.

Construct the interacting flows F_r^R using the same initial random variable X_0 , the same field V , and the same Brownian path as for F_r . Boundedness of the drifts keeps these trajectories in a common compact set on each fixed finite interval, for each realization of (X_0, W) . Local uniform convergence of b_R , the common Lipschitz bound, and Gronwall's inequality imply

$$F_r^R(X_0, W) \longrightarrow F_r(X_0, W)$$

uniformly on compact time intervals, almost surely. Applying the same argument to the differentiated flow equations yields $J_r^R \rightarrow J_r$, with the common bound $\|J_r^R\|_{\text{op}} \leq e^{Kr}$.

For every compact marginal test, the integrand defining the propagated source is bounded in absolute value by

$$\|\nabla\varphi\|_{\infty}e^{Kr}|V(X_0)|.$$

It is integrable and independent of R . Hence the source pairings converge, and the base laws converge weakly. The preceding periodic-kernel estimate, followed by lower semicontinuity, gives (5.5) for b . The initial law and field have remained fixed throughout this last passage, and every quantitative estimate uses only (2.2). This completes the approximation argument.

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